Exercice 3 :

by linearity of $\mathbb{E}$,

. $\mathbb{E}[Y_t] = \mathbb{E}[X_t] + \underbrace{\mathbb{E}[\varepsilon_t]}_{0 \text{ if } \varepsilon_t \text{ is a W.N}}$

$= \mu_X$

where $\mu_X = \mathbb{E}[X_t]$ is a constant of time

. $\forall k \in \mathbb{N}$,

$$\text{Cov}(Y_t, Y_{t-k}) = \text{Cov}(X_t + \varepsilon_t, X_{t-k} + \varepsilon_{t-k})$$

by bi-linearity of Cov, we have

$$\text{Cov}(Y_t, Y_{t-k}) = \text{Cov}(X_t, X_{t-k}) + \underbrace{\text{Cov}(X_t, \varepsilon_{t-k})}_{0 \text{ by independance}} + \underbrace{\text{Cov}(\varepsilon_t, X_{t-k})}_{0 \text{ by independance}} + \text{Cov}(\varepsilon_t, \varepsilon_{t-k})$$

$$\gamma_Y = \text{Cov}(Y_t, Y_{t-k}) \begin{cases} \sigma_X^2 + \sigma_\varepsilon^2, & \text{if } k = 0 \\ \gamma_X(k) & \text{if } k \neq 0 \end{cases}$$

so $\gamma_Y$ is only a function of the lag $k$, we can conclude that $\{Y_t\}$ is stationary by the above proof.